Neanderthal remains may be earliest human ancestor find killed by a spear

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Archaeologists spotted the distinctive injury after excavating skeletal remains of Neanderthals in Shanidar cave, Iraqi Kurdistan. Photograph: Prof Graeme Barker

A smooth round hole in the shoulder blade of a Neanderthal who died more than 70,000 years ago may be the oldest evidence so far of a human ancestor being killed by a wooden spear.

Archaeologists spotted the distinctive injury after excavating skeletal remains of the adult and other Neanderthals from under nearly 8 metres of rock and sediment in the Shanidar cave in Iraqi Kurdistan.

At least five Neanderthals were buried at the same spot over several years, in a depression no larger than a fridge that ran behind a huge boulder near the entrance to the cave. Some of the bodies appear to have been laid on top of others that were still decomposing.

“They’re coming back to the same spot to place bodies over a number of years or several decades and that’s really quite unusual in in the Neanderthal archaeological record,” said Dr Emma Pomeroy, a biological anthropologist at the University of Cambridge. “It suggests there’s some kind of meaningful behaviour in depositing the dead there.”

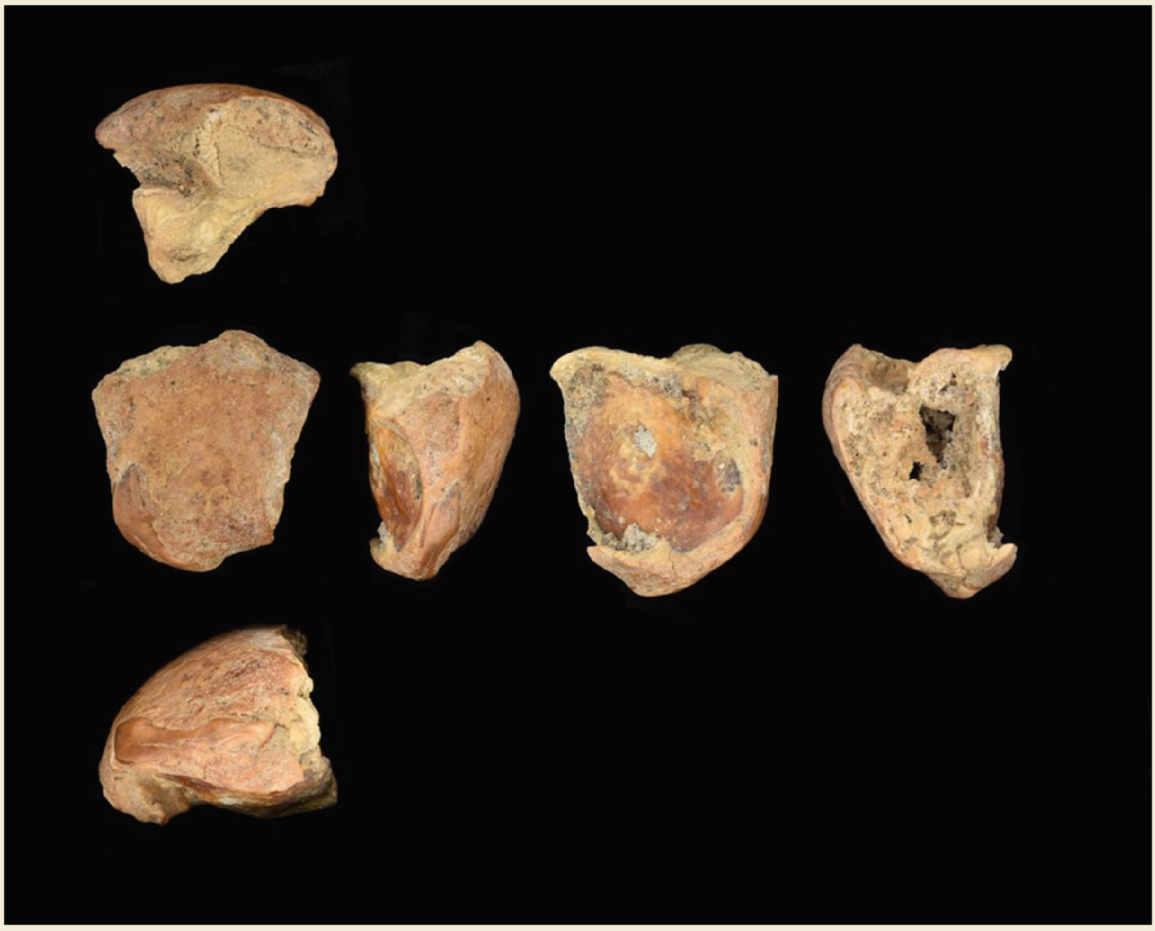
Researchers found only partial remains of the injured Neanderthal, an adult aged about 45. The left shoulder blade that bears the wound was found with seven teeth, several ribs and fragments, and a near complete right hand that was leaning against the boulder.

Close inspection of the shoulder blade revealed three fractures spreading out from the 1.5cm-wide hole and fragments of bone that filled the wound. The shards were probably punched out of the thin bone when the spear hit, the scientists said.

The circular shape of the hole and the pattern of bone damage strongly suggest a wound inflicted by a wooden spear striking the body from the front. But whether it was a hunting accident or an intentional attack – a brutal jab from a foe, perhaps – is impossible to know.

Pomeroy said the position of the wound and the force needed to push the spear through the chest, where it would have punctured the lung and probably severed arteries before punching out of the shoulder blade, pointed to a “kill shot” directed at the heart. “I don’t think that’s something you do accidentally,” she said.

The spear wound was almost certainly lethal, but the individual was in bad shape before the fatal blow. Examination of the teeth revealed a lower incisor split vertically, leaving only half in place, its neighbours twisted inwards, according to details published in Scientific Reports.



Examination found the Neanderthal had a split-tooth caked in hardened plaque. ‘You’d need quite a severe hit in the face to split a tooth like that,’ said Dr Emma Pomeroy from Cambridge University. Photograph: Dr Emma Pomeroy

“You’d need quite a severe hit in the face to split a tooth like that and to rotate the others,” said Pomeroy. “They might well have needed help because they would have been in terrible pain.” The split tooth was caked in hardened plaque that would have taken at least a year to build up.

Chris Hunt, the emeritus professor at Liverpool John Moores University, who co-authored the paper, said Neanderthals at Shanidar were taking ibex as their main prey and could have used wooden spears shaped like modern javelins to kill the animals.

Because plant material decomposes so easily, very few wooden objects survive from prehistory, but timber spears from about 200,000 years ago have been discovered in Germany, preserved in waterlogged ground. The pointed end of a rounded wooden spear discovered in Clacton-on-Sea in 1911 is the oldest known at over 400,000 years old.

Last week, scientists in Spain described hundreds of wooden imprints and artefacts that show Neanderthals embraced woodwork, making trays, plates, tripods, spears and other items. But proof of fatal skirmishes is harder to find in fossils: we may never know how

the wounded Neanderthal at Shanidar died. “Conceivably it was a hunting accident,” Hunt said. “We can’t, however, rule out that purposeful interpersonal violence was involved.”

生词标注（新闻高频 + 词汇第二义项）

词	词性	中文	备注
shoulder blade	n.	肩胛骨	“a smooth round hole in the shoulder blade” 肩胛骨上一个光滑的圆孔；解剖学固定说法，blade 本义“刀片”
evidence	n.	证据	“the oldest evidence so far” 迄今最古老的证据；不可数名词，不能说 an evidence，量词用 a piece of evidence
excavating	v.	发掘、挖出	“after excavating skeletal remains” 发掘出骨骸之后；考古新闻高频动词，名词 excavation 发掘现场
sediment	n.	沉积物	“under nearly 8 metres of rock and sediment” 在近八米厚的岩石与沉积物之下；地质考古常用词
depression	n.	凹坑、洼地	“in a depression no larger than a fridge” 在一个不比冰箱大的凹坑里；此处取地形义，不是常见的“抑郁”
decomposing	v.	腐烂、分解	“others that were still decomposing” 其他尚在腐烂的尸体；植物、遗体分解均可用，名词 decomposition

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record	n.	记录（总体）；档案	“in the Neanderthal archaeological record” 在尼安德特人的考古记录中；此处指某一领域留存的全部材料，非“唱片 / 纪录”
partial	adj.	部分的、残缺的	“only partial remains” 仅存部分遗骸；另义“偏袒的”，阅读时靠上下文分辨
fractures	n.	骨折、裂痕	“three fractures spreading out from the hole” 自孔洞向外延伸的三道裂痕；医学与材料学通用
shards	n.	碎片	“fragments of bone… The shards were probably punched out” 被击出的骨片；常指陶、玻璃、骨的尖锐碎片
inflicted	v.	造成（伤害）、施加	“a wound inflicted by a wooden spear” 由木矛造成的伤口；固定搭配 inflict damage/injury/wound on sb
pointed to	phr.	表明、指向	“pointed to a ‘kill shot’ directed at the heart” 表明那是一记射向心脏的致命一击；point to 的第二义项是“说明、预示”，不是“用手指”
lethal	adj.	致命的	“The spear wound was almost certainly lethal” 这处矛伤几乎肯定是致命的；同义 deadly、fatal
incisor	n.	门齿	“a lower incisor split vertically” 一颗下门齿被竖向劈开；牙科词，incise 切割

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plaque	n.	牙菌斑；（纪念）匾额	“caked in hardened plaque” 结着硬化的牙菌斑；另一常见义是墙上的铭牌，同形不同义
emeritus professor	n.	荣休教授	“the emeritus professor at Liverpool John Moores University” 荣休教授；退休后保留头衔，与 emerita 相对
prey	n.	猎物	“taking ibex as their main prey” 以野山羊为主要猎物；prey on 捕食，易与 pray（祈祷）混淆
waterlogged	adj.	浸水的、水浸的	“preserved in waterlogged ground” 在水浸的土层中保存下来；考古保存有机物的典型环境
rule out	phr.	排除（可能性）	“We can’t, however, rule out that…” 然而我们不能排除……；让步句尾的软化表达，判断题常考
skirmishes	n.	小规模冲突	“proof of fatal skirmishes is harder to find” 致命冲突的证据更难找到；比 battle 规模小，新闻常用来淡化冲突程度